1.1 Number

You need to:

- **Identify and use natural numbers, integers (positive, negative and zero), prime numbers, square and cube numbers, common factors and common multiples, rational and irrational numbers (e.g. π, $\sqrt{2}$), real numbers, reciprocals.**

Natural numbers are the numbers you use to count. So the natural numbers are 1, 2, 3, 4,...

Integers are 'whole numbers'. They can be positive or negative (with zero in between). So the integers are the numbers ...−3, −2, −1, 0, 1, 2, 3,...

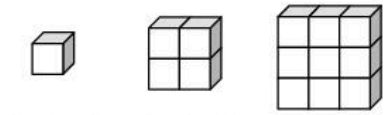
Positive integers are the numbers 1, 2, 3, 4,...

Negative integers are the numbers −1, −2, −3, −4,...

Prime numbers have only two (different) factors (i.e. 1 and itself). So 1 is not a prime number. The prime numbers are 2, 3, 5, 7, 11, 13, 17, 19,... (The number of primes is infinite).

Square numbers

$1^2 = 1$, $2^2 = 4$, $3^2 = 9$, ... So the numbers 1, 4, 9,... are square numbers.



$1 = 1 \times 1$ $4 = 2 \times 2$ $9 = 3 \times 3$

Cube numbers

$1^3 = 1$, $2^3 = 8$, $3^3 = 27$, ... So the numbers 1, 8, 27,... are cube numbers.

Apply

Make a list of:

- the first 15 prime numbers
- the first 15 square numbers
- the first 8 cube numbers.

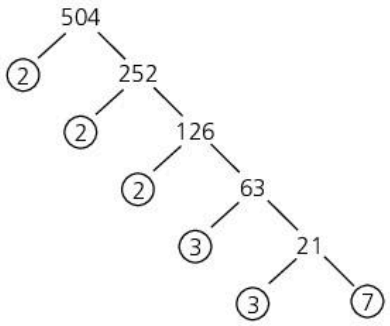
Key skills

You need to be able to write any number as the **product of prime factors.**

Worked example

Express 504 as the product of prime factors. **[2 marks]**

Divide 504 by the smallest possible prime number, in this case 2. Continue until you have only prime numbers in your tree.



Hence $504 = 2^3 \times 3^2 \times 7$

Exam tip

Use a factor tree to help you. Give your answer using index notation.

3 is a **common factor** of 9 and 12 since 3 is a factor of both 9 and 12.

30 is a **common multiple** of 6 and 15 since 30 is a multiple of both 6 and 15.

Worked example

Find the highest common factor and lowest common multiple of 60 and 504. **[4 marks]**

$60 = 2^2 \times 3 \times 5$

$504 = 2^3 \times 3^2 \times 7$

The highest common factor is the product of the prime factors *common* to both numbers, in this case, 2^2 and 3.

$HCF = 2^2 \times 3 = 12$

The lowest common multiple is the product of the largest power of each prime that appears in either number, in this case, 2^3, 3^2, 5 and 7.

$LCM = 2^3 \times 3^2 \times 5 \times 7 = 2520$

Key skills

You need to be able to work out the **Highest Common Factor** (**HCF**) and the **Lowest Common Multiple** (**LCM**) of two numbers.

Exam tip

Write both numbers as the product of prime factors first. 2^2 is common to both numbers, but not 2^3. Likewise, 3 is common to both numbers, but not 3^2.

A **rational number** is a number which can be expressed in the form $\frac{p}{q}$ where p and q are whole numbers.

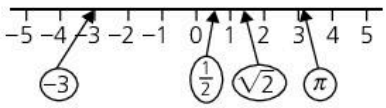
- All decimals which recur are rational. For example, $0.\dot{3} = 0.333333...$ is rational since $0.\dot{3} = \frac{1}{3}$
- All decimals which terminate (i.e. which end) are rational. For example, 0.625 is a rational number since 0.625 can be written as $\frac{625}{1000} = \frac{5}{8}$

Watch out!

$\sqrt{2\frac{1}{4}}$ doesn't look rational but it is since $\sqrt{2\frac{1}{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2}$

An **irrational number** is a number which cannot be expressed in the form $\frac{p}{q}$ where p and q are integers. For example, π, $\sqrt{2}$ and $\sqrt[3]{7}$ are all irrational numbers.

A **real number** is any rational or irrational number that can be represented on a number line.



Exam tip
Look for square numbers in the list and work out the square roots of these first.

Worked example

$\sqrt{4}$ $\sqrt{15}$ $\sqrt{25}$ $\sqrt{36}$ $\sqrt{144}$

From the list above, write down:

(a) an odd prime number
(b) two factors of 42
(c) an irrational number
(d) a multiple of 4. **[4 marks]**

(a) $\sqrt{25} = 5$ which is an odd prime number
(b) $\sqrt{4} = 2$ and $\sqrt{36} = 6$, and both 2 and 6 are factors of 42
(c) $\sqrt{15}$ is an irrational number
(d) $\sqrt{144} = 12$ which is a multiple of 4

The **reciprocal** of a number is 'one over' the number. For example, the reciprocal of 4 is $\frac{1}{4}$ and the reciprocal of n is $\frac{1}{n}$.
If you are asked to find the reciprocal of a fraction, you just 'flip it over'. For example, the reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Questions

1 Which of these numbers is **not** a rational number?

$\frac{3}{5}$ $\sqrt{7}$ 0.6 $-1\frac{2}{3}$ $\sqrt{25}$

2 An integer n is such that $80 \leq n < 90$.
Write down a value of n which is:
a a multiple of both 3 and 4
b a prime number
c a factor of 1700.

3 Find:
a a prime number which is a factor of 49
b an even prime number.

4 **a** Write down all the factors of 21.
b Write down all the factors of 28.
c Hence find the HCF of 21 and 28.

5 $\sqrt{9}$ $\sqrt{15}$ $\sqrt{49}$ $\sqrt{16}$ $\sqrt{64}$ $\sqrt{121}$
From the list above, write down:
a a prime number less than 10
b a factor of 22
c a power of 2
d an irrational number.

6 Write each number as the product of prime factors (e.g. $72 = 2^3 \times 3^2$).
a 30 **b** 24 **c** 18
d 28 **e** 105 **f** 64

7 Use question **6** to find the highest common factor of each pair of numbers.
a 30 and 24
b 28 and 64
c 18 and 105

8 Use question **6** to find the lowest common multiple of each pair of numbers.
a 30 and 64
b 24 and 105
c 28 and 18

9 Write down the reciprocal of the following numbers:
a 5 **b** $\frac{1}{3}$ **c** $\frac{4}{5}$ **d** $\frac{17}{13}$

To **Raise your grade** now try question 3, page 49